

Appendix: Seeing the Rare: Meta-Aware Pointer Networks for Long-Tailed Video Captioning

I. COMPARISON ON LARGE SCALE DATASETS.

A. ActivityNet and YouCook2 Dataset

We extend the evaluation to larger and more diverse benchmarks for completeness. ActivityNet and YouCook2 are originally designed for dense video captioning, where multiple temporally localized captions are provided per video. In contrast, our focus is on generating a single holistic caption, consistent with widely used datasets.

To adapt these datasets, we treat each video as a whole and generate one global caption. Although this reduces fine-grained temporal details, it enables evaluation on open-domain, large-scale datasets covering diverse activities. The comparative results in Table I show that MAPNet consistently outperforms the baseline VSVCap, demonstrating strong generalization and scalability to complex video domains.

Table I: Comparison of MAPNet on ActivityNet and YouCook2 dataset for single-sentence video captioning.

Dataset	Model	BLEU-4	METEOR	ROUGE-L	CIDEr
ActivityNet	VSVCap	8.7	12.1	27.4	16.3
	MAPNet	10.9	13.8	29.6	21.7
YouCook2	VSVCap	9.5	13.2	28.1	18.9
	MAPNet	11.8	15.0	30.4	23.5

Table II: Comparison of MAPNet on VATEX dataset.

Methods	BLEU-4	METEOR	ROUGE-L	CIDEr
ORG-TRL [1]	32.1	22.2	48.9	49.7
VSVCap [2]	39.6	26.5	50.3	70.2
SySCapNet [3]	40.1	26.9	53.0	71.5
MAPNet	42.7	29.4	61.57	71.2

B. VATEX Dataset

To broaden the evaluation, we test on the large-scale VATEX benchmark, which is more diverse and challenging than MSVD and MSR-VTT due to its scale and bilingual annotations. As shown in Table II, performance scores are naturally lower, reflecting the dataset’s lexical variability. Still, MAPNet achieves strong BLEU scores by accurately reproducing key visual entities, competitive METEOR scores through improved generalization, and reasonable CIDEr scores despite reduced overlap from diverse references. These results confirm the robustness of MAPNet on large-scale, open-domain datasets.

II. SYSTEMATIC QUALITATIVE ERROR ANALYSIS

We have expanded the qualitative analysis by conducting a systematic error study on 40 randomly sampled videos from MSR-VTT and MSVD, stratified by rare and head-class terms. Each MAPNet caption was labeled by three annotators using

a controlled taxonomy of error types: Missing Rare Action (MRA), Wrong Rare Entity (WRE), Overly Generic (OG), Hallucination (H), Repetition (R), Temporal Error (TE), as shown in Figure 1. The most frequent issues were MRA (21%) and WRE (18%), showing that rare-class modeling remains the most challenging aspect.

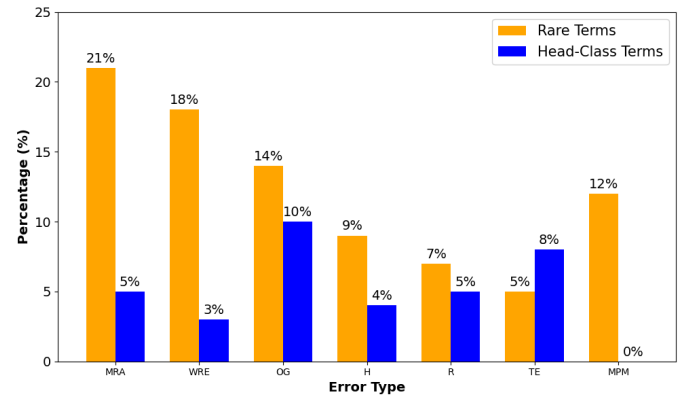


Figure 1: Systematic Qualitative Error Analysis of MAPNet.

III. CALCULATION OF COMPUTATION OVERHEAD

Table III summarizes the computational cost of different variants of MAPNet. The pointer network introduces a small parameter and inference-cost increase ($\approx 2 - 3\%$), while the meta-learning component increases per-epoch training time ($\approx 17\%$ in this illustrative example) due to inner-loop updates. Importantly, meta-learning reduces the number of epochs needed to reach the target validation CIDEr ($22 \rightarrow 18$).

Table III: Illustrative computation overhead: Baseline (VSVCap) vs. Pointer (P) vs. Meta (M) vs. Meta+Pointer (M+P).

Metric	VSVCap	+P	+M	M+P	Overhead
Parameters (M)	42.0	43.1	42.0	43.1	+2.6%
FLOPs(GFLOPs/sample)	8.40	8.60	8.40	8.60	+2.4%
Training time/epoch	9.20	9.50	10.60	10.80	+17.4%

REFERENCES

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